

# From L<sup>A</sup>T<sub>E</sub>X to Quarto

## Scientific Writing, Reproducible Publishing, and Beyond

Imad El Badisy, PhD

2026-06-24

# Workshop Outline I

1. Part 1 - Scientific Writing with  $\text{\LaTeX}$
2. Part 2 - Quarto for Reproducible Publishing

## Section 1

### 1. Part 1 - Scientific Writing with $\text{\LaTeX}$

## 1.1 A Brief History

### **T<sub>E</sub>X - D. Knuth, 1978**

- ▶ Underlying engine, still used today
- ▶ Deterministic, reproducible output

### **L<sup>A</sup>T<sub>E</sub>X - L. Lamport, 1984**

- ▶ Macro layer on top of T<sub>E</sub>X
- ▶ Separates content from formatting
- ▶ Became the standard for scientific writing

| Year | Event                                       |
|------|---------------------------------------------|
| 1978 | T <sub>E</sub> X released                   |
| 1984 | L <sup>A</sup> T <sub>E</sub> X 2.09        |
| 1994 | L <sup>A</sup> T <sub>E</sub> X2e (current) |
| 1999 | pdfT <sub>E</sub> X → direct PDF            |
| 2004 | XeT <sub>E</sub> X → Unicode                |
| 2007 | LuaT <sub>E</sub> X → scripting             |
| 2012 | Overleaf launched                           |

## 1.2 L<sup>A</sup>T<sub>E</sub>X is not Overleaf!

For many students, **Overleaf** = **LaTeX** but they are not the same thing!



### L<sup>A</sup>T<sub>E</sub>X (1984)

- ▶ A typesetting *language + compiler*
- ▶ Runs anywhere: machine, server, cloud
- ▶ Free and open source
- ▶ Works offline in any editor

### Overleaf (2012)

- ▶ A *web app* built on top of L<sup>A</sup>T<sub>E</sub>X
- ▶ Runs the compiler on their servers
- ▶ Adds: browser UI, collaboration, cloud storage
- ▶ You are still writing L<sup>A</sup>T<sub>E</sub>X

**Analogy:** L<sup>A</sup>T<sub>E</sub>X is the engine. Overleaf is the car. Swap the car - the engine stays the same.

## 1.3 Why $\text{\LaTeX}$ ?

### The problem with Word

- ▶ Copy-paste figures from R  $\rightarrow$  manual update on every change
- ▶ `final_v3_FINAL_revised.docx`
- ▶ Math typesetting is painful

| Feature           | Word    | $\text{\LaTeX}$ |
|-------------------|---------|-----------------|
| Typesetting       | Medium  | Excellent       |
| Math              | Poor    | Native          |
| Figure placement  | Manual  | Algorithmic     |
| Bibliography      | Plugin  | Built-in        |
| Version control   | Hard    | Easy            |
| Journal templates | Limited | Extensive       |

## 1.4 Where to Write $\text{\LaTeX}$ - Platforms

| Platform                        | Type    | Best for               |
|---------------------------------|---------|------------------------|
| <b>Prism</b> (OpenAI)           | Browser | Beginners + AI writing |
| <b>Overleaf</b>                 | Browser | Collaboration          |
| <b>TeXStudio</b>                | Desktop | Full-featured editor   |
| <b>VS Code</b> + LaTeX Workshop | Desktop | Programmers / Git      |
| <b>TeXmaker</b>                 | Desktop | Lightweight editor     |
| <b>RStudio</b>                  | Desktop | R users                |

**Recommendation:** beginners  $\rightarrow$  Prism or Overleaf; advanced  $\rightarrow$  VS Code + Git.

## 1.5 Why Document Classes?

`\documentclass` is the first decision - it sets the layout, section hierarchy, and all defaults.

| Class      | Use                      |                                                                                      |
|------------|--------------------------|--------------------------------------------------------------------------------------|
| article    | Papers, short reports    | <code>% Generic paper</code><br><code>\documentclass[12pt]{article}</code>           |
| report     | Theses, longer reports   | <code>% Two-column journal</code><br><code>\documentclass[twocolumn]{article}</code> |
| book       | Books, front/back matter | <code>% Elsevier submission</code><br><code>\documentclass{elsarticle}</code>        |
| beamer     | Slides                   | <code>% Slide deck</code><br><code>\documentclass{beamer}</code>                     |
| elsarticle | Elsevier journals        | <code>\usepackage{Madrid}</code>                                                     |
| IEEEtran   | IEEE journals            |                                                                                      |

**Why it matters:** `\section{}` renders differently across classes. Always start from the journal's `.cls` file.

The class file controls everything - margins, fonts, title format, header/footer.

## 1.6 Document Structure

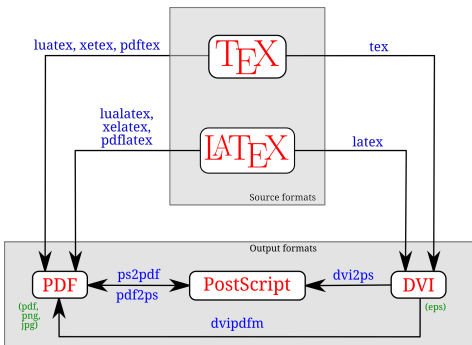
```
\documentclass[12pt, a4paper]{article}    % (1) class

% (2) Preamble - packages and settings
\usepackage{amsmath, amssymb}           % math
\usepackage{graphicx}                   % figures
\usepackage{booktabs}                   % tables
\usepackage{hyperref}                   % links
\usepackage[style=apa, backend=biber]{biblatex}
\addbibresource{references.bib}

\title{My Paper}
\author{Author Name}
\date{\today}

% (3) Body
\begin{document}
\maketitle
\tableofcontents
```

## 1.7 How to Compile



```
# pdflatex (most common)
pdflatex paper.tex
biber paper          # bibliography
pdflatex paper.tex # 2nd pass:
```

```
# xelatex (Unicode + fonts)
xelatex paper.tex
```

```
# latexmk (recommended)
# automates all passes
latexmk -xelatex paper.tex
```

**Run at least twice** -  
cross-references are resolved on the  
second pass.

**Overleaf** handles this  
automatically.

## 1.8 Mathematical Notation

**Inline:**  $\hat{\beta} = 0.42 \rightarrow$

$\hat{\beta} = 0.42$

**Display:**

```
\begin{equation}
\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y
\end{equation}
```

**Aligned:**

```
\begin{align}
\ell(\theta) &= \sum_{i=1}^n \log f(y_i | \theta) \\
\hat{\theta} &= \arg\max_{\theta} \ell(\theta)
\end{align}
```

**Custom commands** (define once):

```
\newcommand{\E}{\mathbb{E}}
\newcommand{\Var}{\mathrm{Var}}
\newcommand{\iid}{\overset{iid}{\sim}}
```

| Symbol          | L <sup>A</sup> T <sub>E</sub> X |
|-----------------|---------------------------------|
| $\sum$          | <code>\sum</code>               |
| $\prod$         | <code>\prod</code>              |
| $\int$          | <code>\int</code>               |
| $\hat{\theta}$  | <code>\hat{\theta}</code>       |
| $\alpha, \beta$ | <code>\alpha, \beta</code>      |
| $\mathbb{E}[X]$ | <code>\mathbb{E}[X]</code>      |
| $\leq, \geq$    | <code>\leq, \geq</code>         |
| $\in, \notin$   | <code>\in, \notin</code>        |
| $\sim$          | <code>\sim</code>               |
| $X^{\top}$      | <code>X^{\top}</code>           |

**Demo**  $\rightarrow$

[demos/latex/02-math/](#)

# 1.9 Diagrams - TikZ and Alternatives

**TikZ** - standard  $\text{\LaTeX}$  diagram package

```
\usepackage{tikz}
\usetikzlibrary{arrows.meta, positioning}

\begin{tikzpicture}[node distance=1cm,
  box/.style={draw, rounded corners,
    fill=blue!8, minimum width=2.8cm}]
  \node[box] (A) {Inclusion};
  \node[box, below=of A] (B) {Eligible};
  \node[box, below=of B] (C) {Analysis};
  \draw[->] (A)--(B); \draw[->] (B)--(C);
\end{tikzpicture}
```

Other uses: DAGs, survival trees, neural nets, timelines.

## 1.10 Debugging with AI Coding Agents

Modern AI agents (Claude, Codex, GitHub Copilot, Cursor) accelerate  $\text{\LaTeX}$  and Quarto debugging.

### What to give the agent:

1. The **exact error message** from the terminal or log
2. The **minimal code block** that triggers the error
3. The **context**: document class, relevant packages, what you expected

### Effective prompt:

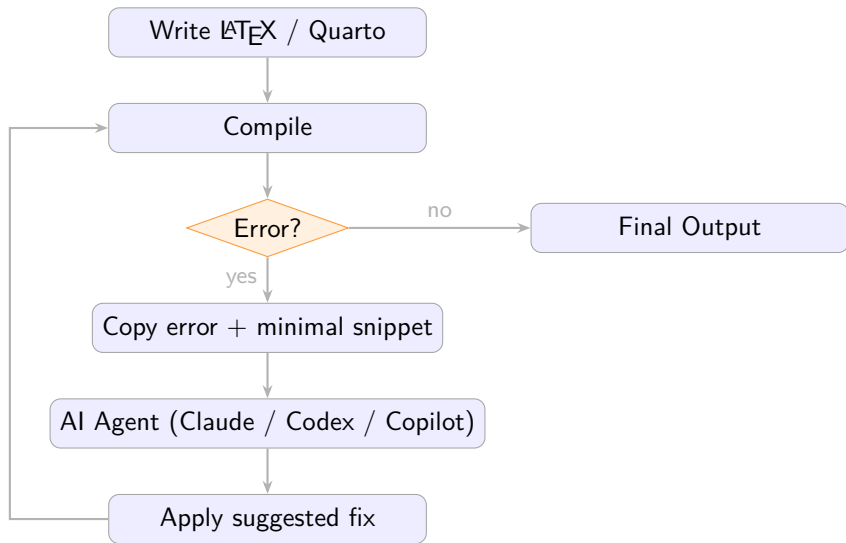
I am writing  $\text{\LaTeX}$  with `\documentclass{article}` and `\usepackage{}`  
The following snippet fails to compile:

```
\begin{align}  
  y = x^2  
\end{align*}
```

Error: "Extra `\begin{align}`."

What is wrong and how do I fix it?

## 1.11 AI-Assisted Debugging Workflow



**Demo** → demos/latex/ - 01-minimal, 02-math, 03-template, 04-bibliography

## Section 2

### 2. Part 2 - Quarto for Reproducible Publishing

## 2.1 From $\text{\LaTeX}$ to Quarto - Why?

| $\text{\LaTeX}$              | Quarto                                 |
|------------------------------|----------------------------------------|
| Excellent typesetting        | Same $\text{\LaTeX}$ engine underneath |
| Manual copy-paste of results | Results computed inside the document   |
| One output format            | HTML, PDF, slides, website, book       |
| Text only                    | Code + text unified                    |
| Steep learning curve         | Markdown + YAML + code                 |

**Quarto =  $\text{\LaTeX}$  quality + reproducible code + multiple outputs  
from one source**

## 2.2 What Quarto Solves

Quarto turns one plain-text `.qmd` source into many publishable outputs.

| Need                    | Output format                        |
|-------------------------|--------------------------------------|
| Web report / dashboard  | HTML                                 |
| Submission / print      | PDF (via $\text{\LaTeX}$ )           |
| Lecture slides          | RevealJS (browser) or Beamer (PDF)   |
| Course or research site | Website                              |
| Thesis / textbook       | Book $\rightarrow$ HTML + PDF + EPUB |
| Repeated journal format | Template / extension                 |

**One source file  $\rightarrow$  many targets. No copy-paste.**

## 2.3 Compilers and Install

### Install check:

```
quarto --version  
quarto check          # diagnose engines
```

### Render commands:

```
quarto render report.qmd  
quarto render report.qmd --to pdf  
quarto render report.qmd --to html  
quarto preview report.qmd # live
```

`quarto check` shows which engines are installed and working.

## 2.4 File Anatomy

Every .qmd file has three layers:

```
---  
title: "My Report"          # YAML  
author: "Student Name"  
format:  
  html:  
    toc: true  
    code-fold: true  
  pdf:  
    number-sections: true  
execute:  
  warning: false  
bibliography: refs.bib  
---
```

```
## Introduction      ← Markdown block
```

```
This is bold and italic.
```

- First item
- Second item

```
```{r}                ← Code chunk
```

```
#| label: fig-km  
#| fig-cap: "KM curve."  
survfit(...) |> plot()  
```
```

Rule: write once → render to any format.

## 2.5 Markdown - The Prose Layer

**Markdown** (J. Gruber, 2004) - lightweight markup, readable as plain text, converts to HTML/PDF via Pandoc.

- ▶ In Quarto: Markdown is the prose layer.  $\text{\LaTeX}$  runs underneath for PDF.

### Syntax comparison

| Element | $\text{\LaTeX}$               | Markdown               |
|---------|-------------------------------|------------------------|
| Bold    | <code>\textbf{x}</code>       | <code>**x**</code>     |
| Italic  | <code>\textit{x}</code>       | <code>*x*</code>       |
| Section | <code>\section{T}</code>      | <code>## T</code>      |
| Bullet  | <code>\begin{itemize}</code>  | <code>- item</code>    |
| Link    | <code>\href{u}{t}</code>      | <code>[t](u)</code>    |
| Code    | <code>\texttt{x}</code>       | <code>`x`</code>       |
| Image   | <code>\includegraphics</code> | <code></code> |

## 2.6 YAML - The Control Layer

YAML controls the entire output without touching the content.

```
---
title: "Survival Analysis Report"
author: "Imad El Badisy"
date: today
format:
  html:
    toc: true
    theme: cosmo
  pdf:
    toc: true
    number-sections: true
    pdf-engine: xelatex
execute:
  message: false
bibliography: references.bib
---
```

Change the format block to completely different output, same as before

## 2.7 Code Chunks and Inline Code

**Chunk options** (`#|` syntax - same for R, Python, Julia):

| Option               | Effect                                   |
|----------------------|------------------------------------------|
| <code>label</code>   | names output for <code>@fig-</code> refs |
| <code>echo</code>    | show / hide source code                  |
| <code>eval</code>    | run / skip chunk                         |
| <code>fig-cap</code> | figure caption                           |
| <code>tbl-cap</code> | table caption                            |
| <code>warning</code> | show / hide warnings                     |

## 2.8 Cross-References and Citations

**Figures, tables, equations** - use label prefixes:

```
! [Flow diagram] (flow.png) {#fig-flow}      → cite as @fig-flow  
: Patient groups {#tbl-groups}                → cite as @tbl-groups  
$$  $h(t|x) = h_0(t)e^{\{x^{\top}\beta\}}$  $$ {#eq-cox} → @eq-cox
```

**Citations** - add .bib in YAML, then cite inline:

```
bibliography: references.bib
```

The Cox model [@cox1972] is widely used in survival analysis.

**Label prefixes are mandatory:** fig-, tbl-, eq-, sec- - wrong prefix = broken reference.

**Demo** → [demos/quarto/02-crossrefs/](#)

## 2.9 Output Formats

| Format          | YAML key        | Render command | Best for               |
|-----------------|-----------------|----------------|------------------------|
| HTML            | html            | --to html      | Web, interactive       |
| PDF             | pdf             | --to pdf       | Submission, archiving  |
| Word            | docx            | --to docx      | Collaborators          |
| Beamer slides   | beamer          | --to beamer    | PDF lectures           |
| RevealJS slides | revealjs        | --to revealjs  | Browser talks          |
| Website         | project website | quarto preview | Course / research site |
| Book            | project book    | quarto render  | Thesis, textbook       |
| EPUB            | epub            | --to epub      | Mobile reading         |

All formats share one source - switching is a YAML change, not a rewrite.

## 2.10 Slides - RevealJS vs Beamer

### RevealJS - browser slides

```
format:
  revealjs:
    theme: default
    slide-number: true
    incremental: true
```

- ▶ Runs in any browser
- ▶ Animations, fragments, speaker notes
- ▶ Easy to share: just a URL
- ▶ Interactive elements (Observable)

**Structure:** with `slide-level: 2`, # headings → sections, ## headings → slides.

**Demo** → `demos/quarto/03-slides/` and `demos/quarto-lecture/02-slides/`

### Beamer - PDF slides

```
format:
  beamer:
    theme: Madrid
    colortheme: crane
    pdf-engine: xelatex
```

- ▶ Classic conference / print format
- ▶ Full  $\text{\LaTeX}$  math and TikZ
- ▶ Works offline, always renders the same

## 2.11 Projects and `_quarto.yml`

Use a **project** when you have more than one file.

```
my-project/  
  _quarto.yml  
  index.qmd  
  methods.qmd  
  results.qmd  
  references.bib  
  figures/
```

```
quarto render      # all files  
quarto preview     # live reload  
quarto create project book  
quarto create project website
```

```
# _quarto.yml - shared config  
project:  
  type: website      # or: book, document  
  output-dir: _output  
website:  
  title: "My Course"  
  navbar:  
    left:  
      - index.qmd  
      - methods.qmd  
format:  
  html:  
    toc: true  
    theme: cosmo  
  pdf:  
    number-sections: true
```

## 2.12 Templates

Use templates when the same structure repeats across documents.

### Apply a template:

```
quarto use template owner/repository
```

### Workflow:

1. Render unchanged - verify it compiles
2. Edit YAML (title, author, date)
3. Render again
4. Replace body content
5. Final render

### Useful repos:

- ▶ `quarto-journals` - journal formats
- ▶ `quarto-thesis` - thesis template
- ▶ `quarto-revealjs-clean` - clean slides

### Create your own template:

```
my-template/  
  template.qmd      ← example document  
  _extensions/  
    my-template/  
      _extension.yml  
      custom.tex    ← LaTeX preprocessor  
      theme.scss     ← HTML theme
```

A template is just a `.qmd` file with a custom format - share it with `quarto use template`.

### Demo →

`demos/quarto-lecture/05-templates`

## 2.13 Lab - Build Your Own Templates

**3 hours remaining - build from scratch in pairs or solo:**

| Task                             | Output             | Key concepts                         |
|----------------------------------|--------------------|--------------------------------------|
| <b>1</b> - Minimal report        | HTML + PDF         | YAML, Markdown, code chunk           |
| <b>2</b> - Slide deck            | RevealJS or Beamer | slide-level, themes                  |
| <b>3</b> - Website               | HTML site          | _quarto.yml, navbar                  |
| <b>4</b> - Book / thesis chapter | HTML + PDF         | chapters, cross-refs                 |
| <b>5</b> - eBook                 | EPUB               | epub format, covers                  |
| <b>6</b> - Custom template       | reusable format    | _extensions/, custom $\text{\LaTeX}$ |

**Start from demos/quarto-lecture/ - one folder per task.**